Introduction to Computing

Lecture 13

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Memory Address, References and Pointers

Getting Memory Address

When a variable is created in C++, a memory address is assigned to the variable. And when we assign a value to the variable, it is stored in this memory address.

To access it, use the **&** operator, and the result will represent where the variable is stored:

Example:

```
int a = 2;

cout << &a; // Outputs 0x6dfed4
```

Reference Variable

A reference variable is a "reference" to an existing variable, and it is created with the & operator:

```
int a = 2;
int &b= a; // reference to a
```

- The reference variable can only be initialized at the time of its creation
- The reference variable returns the address of the variable preceded by the reference sign ‘&’
- The reference variable can never be reinitialized again in the program
- The reference variable can never refer to NULL

Pointer Variable

A pointer is a variable that stores the memory address as its value.

- A pointer variable points to a data type of the same type
- It is created with the * operator.
- The address of the variable you're working with is assigned to the pointer

Example:

```
int *a ;
int b = 2;
a= &b; // a stores the address of b
```

Accessing Memory Address and Value using Pointer Variable

- Pointer variable holds the address of a variable, so its not a problem
- We can also get the value of the variable through pointer, by using the * operator (**the dereference operator**).
- We can also change the value of the variable by using the * operator

Example:

```
int *a ;
int b = 2;
a= &b;          // a stores the address of b
cout << *a;     // using dereference operator we get value of 'b'
*a = 3;         // using dereference operator we set value of 'b'
cout << b;      // we get 3
```

Accessing Memory Address and Value using Pointer Variable

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Exa

```
int
int
a= &
cout << *a;     // using dereference operator we get value of 'b'
*a = 3;         // using dereference operator we set value of 'b'
cout << b;      // we get 3
```

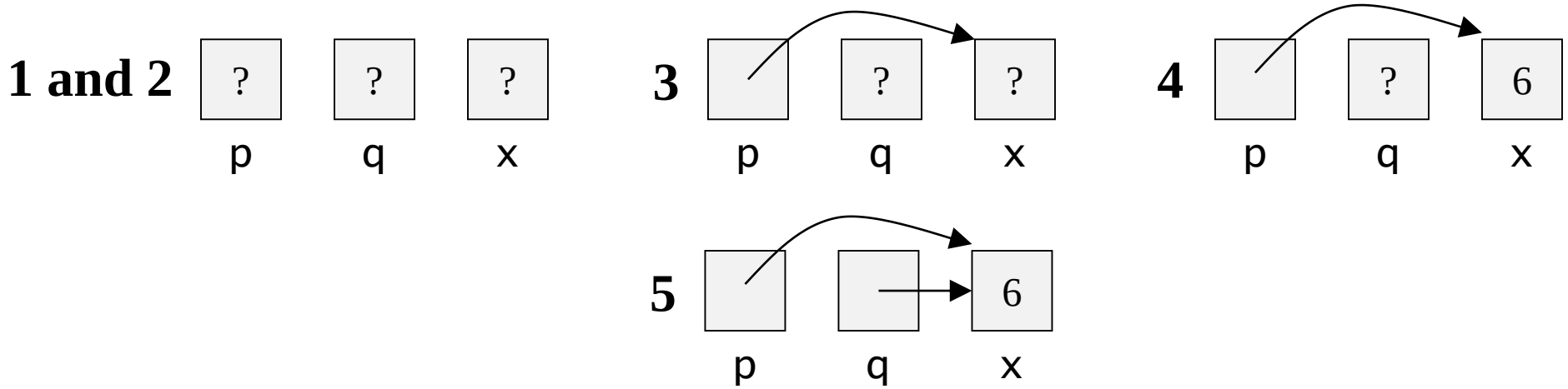
Note:

The * sign can be confusing here, as it does two different things in our code:

- When used in **declaration** (string* ptr), it creates a pointer variable.
- When **not used in declaration**, it act as a dereference operator.

Pointers (Recap)

1.	Pointer variables	int *p, *q;
2.	Static allocation	int x;
3.	Address-of operator	p = &x;
4.	Memory cell to which P points	*p = 6;
5.	Pointer operations	q = p;



“Pass by Value” and “Pass by Reference”

Pass by Value:

- Makes a copy in memory of the actual parameters

- Use pass by value when you are only **using** the parameter for some computation, not changing it

Pass by Reference:

- Forwards the actual parameters

- Use pass by reference when you are **changing** the parameter passed in the program

“Pass by Value”

```
#include <iostream>
using namespace std;

int add(int a)
{
  int b = 0;
  a = a + 1;
  b=a;

  return b;
}


int main() {
  int x = 0;
  int result = add(x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

“Pass by Reference”

```
#include <iostream>
using namespace std;

int add(int* a)
{
  int b = 0;
  *a = *a + 1;
  b=*a;

  return b;
}


int main() {
  int x = 0;
  int result = add(&x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

"Pass by Value"

```
#include <iostream>
using namespace std;

int add(int a)
{
  int b = 0;
  a = a + 1;
  b=a;

  return b;
}


int main() {
  int x = 0;
  int result = add(x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

Function Declaration

"Pass by Reference"

```
#include <iostream>
using namespace std;

int add(int* a)
{
  int b = 0;
  *a = *a + 1;
  b=*a;

  return b;
}


int main() {
  int x = 0;
  int result = add(&x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

Function Declaration

"Pass by Value"

```
#include <iostream>
using namespace std;

int add(int a)
{
  int b = 0;
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  return b;
}

int main() {
  int x = 0;
  int result = add(x);
  cout << result << endl;
  cout << x << endl;
return 0;
}
```

Function Definition

"Pass by Reference"

```
#include <iostream>
using namespace std;

int add(int* a)
{
  int b = 0;
  *a = *a + 1;
  b=*a;

  return b;
}

int main() {
  int x = 0;
  int result = add(&x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

Function Definition

"Pass by Value"

Pass by Pointer ~~"Pass by Reference"~~

```
#include <iostream>
using namespace std;

int add(int a)
{
  int b = 0;
  a = a + 1;
  b=a;

  return b;
}


int main() {
  int x = 0;
  int result = add(x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

Function Calling

```
#include <iostream>
using namespace std;

int add(int* a)
{
  int b = 0;
  *a = *a + 1;
  b=*a;

  return b;
}


int main() {
  int x = 0;
  int result = add(&x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

Function Calling

Another way for “Pass by Reference”

```
#include <iostream>
using namespace std;

int add(int &a)
{
  int b = 0;
  a = a + 1;
  b=a;

  return b;
}


int main() {
  int x = 0;
  int result = add(x);
  cout << result << endl;
  cout << x << endl;
  return 0;
}
```

Reference Variable:
Reference variable is an alias for a variable which is assigned to it.

Different from pointer:

- The reference variable can only be initialized at the time of its creation
- The reference variable returns the address of the variable preceded by the reference sign ‘&’
- The reference variable can never be reinitialized again in the program
- The reference variable can never refer to NULL